

NOVAC Internship – Task 7 Daily Documentation

Task: Conversational Semantic Retrieval Chatbot, Progressive Chunk Retrieval, Whisper Voice Search Integration, and Messaging-Style React Chat Interface

Overview

Today's work focused on transforming the semantic search application into a conversational AI chatbot system capable of progressive semantic retrieval, voice-enabled interaction, and messaging-style communication workflows.

The project evolved from a standard semantic search interface into a conversational retrieval platform where users can interact naturally with uploaded documents through chatbot-based semantic querying.

The implementation introduced:

- progressive chunk retrieval
- conversational semantic memory
- chatbot-style UI rendering
- Whisper voice search integration
- greeting interaction workflows
- message-based frontend interaction
- retrieval continuity handling

By the end of the session, the system was successfully able to:

- perform conversational semantic retrieval
- return highest-ranked semantic chunks progressively
- support "more", "continue", and follow-up retrieval workflows
- support microphone-based voice input
- transcribe speech using Whisper
- display chatbot-style message bubbles
- maintain conversational retrieval memory
- support greeting-based interaction handling

Resources and References Used

- React Documentation
- FastAPI Documentation
- MongoDB Documentation
- Sentence Transformers Documentation
- OpenAI Whisper Documentation
- ffmpeg Documentation
- MediaRecorder API Documentation
- JavaScript Fetch API Documentation

Concepts Learned

Conversational Semantic Retrieval

Learned how semantic search systems can maintain conversational memory.

Understood:

- ranked semantic retrieval
- progressive response delivery
- conversational chunk continuation
- retrieval memory handling
- semantic similarity ranking

Implemented:

- next-best chunk retrieval
- conversational follow-up handling
- progressive semantic response flow

Conversational Memory Handling

Learned how backend systems maintain temporary conversational state.

Implemented:

- last semantic results storage
- retrieval index tracking
- conversational continuity
- follow-up query handling

Understood:

- backend session-style retrieval memory
- conversational indexing workflows
- retrieval progression systems

Whisper Voice Search Integration

Learned how speech-to-text systems integrate into AI chatbot workflows.

Implemented:

- microphone access
- audio recording workflows
- MediaRecorder integration
- audio upload handling
- Whisper transcription API integration

Understood:

- browser microphone permissions
- audio blob generation
- speech-to-text pipelines
- Whisper backend communication

Chatbot-Based UI Design

Learned how modern AI chatbot interfaces are structured.

Implemented:

- messaging-style conversation UI
- left-right message alignment
- user message rendering
- bot response rendering
- conversational flow styling

Understood:

- conversational UI architecture
- dynamic message rendering
- React state-based conversation management

Greeting Interaction Workflows

Learned how conversational AI systems handle non-search interactions.

Implemented:

- greeting detection
- conversational greeting responses
- startup welcome message
- conversational entry workflow

Supported greetings:

- hi
- hello
- hey
- good morning
- good evening

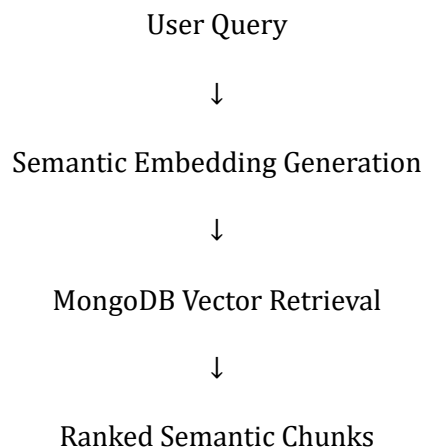
Libraries, Modules, and Components Learned

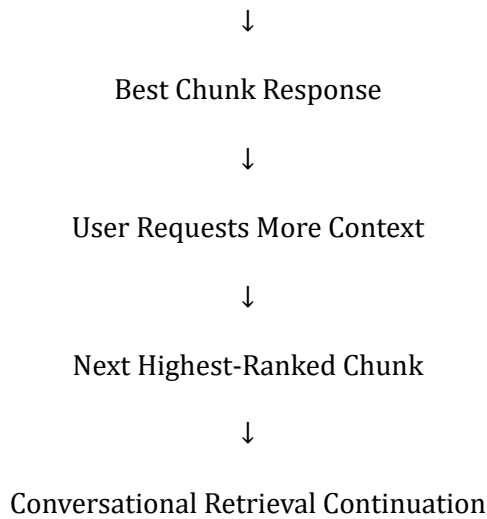
- **React.js:** Used for chatbot frontend rendering and conversational UI.
- **FastAPI:** Used for backend API routing and retrieval handling.
- **MongoDB:** Used for semantic chunk storage.
- **Sentence Transformers:** Used for semantic embedding generation.
- **Whisper:** Used for speech-to-text transcription.
- **MediaRecorder API:** Used for browser audio recording.
- **ffmpeg:** Used for audio processing support required by Whisper.

Practical Implementation

- Converted semantic search into conversational retrieval workflow
- Implemented progressive chunk retrieval logic
- Added conversational retrieval memory
- Added “more” and “continue” retrieval handling
- Added chatbot-style message rendering
- Added Whisper voice recording integration
- Added microphone recording workflows
- Added speech-to-text API integration
- Added conversational greeting handling
- Added startup welcome message
- Improved chatbot conversational experience
- Added voice query auto-fill workflow

Architecture Understood





Challenges Faced

Conversational Retrieval Logic

Encountered issues where:

- follow-up queries reset retrieval flow
- retrieval memory was not preserved
- semantic continuation failed

Resolved by implementing:

- retrieval result storage
- conversational indexing
- progressive chunk handling

React State Rendering Issues

Faced problems where:

- chatbot messages disappeared
- conversational rendering broke
- UI state updates failed

Resolved through:

- proper React state updates
- dynamic message array handling
- conversational rendering corrections

Whisper Integration Issues

Encountered:

- installation failures
- dependency issues
- ffmpeg configuration problems
- microphone permission handling issues

Successfully resolved and completed:

- Whisper installation
- ffmpeg setup

- audio transcription workflows

Voice UI Overwrite Issues

During frontend restructuring:

- microphone button disappeared
- conversational UI overwrote previous components

Resolved by:

- re-integrating voice workflows
- restoring recording state management
- reconnecting Whisper API integration

Conclusion

Today's work significantly improved the system from a standard semantic search interface into a conversational AI document assistant capable of progressive retrieval workflows and voice-enabled interaction.

The platform now supports:

- conversational semantic retrieval
- progressive chunk continuation
- chatbot-based interaction
- voice-enabled querying
- conversational UI workflows
- retrieval memory handling
- greeting interaction systems
- speech-to-text integration

The concepts learned today form the foundation of:

- conversational RAG systems
- enterprise semantic assistants
- AI-powered document chatbots
- voice-enabled retrieval systems
- conversational AI interfaces
- intelligent retrieval workflows

Prepared By

Divesh Kumar S